higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N5

(8090075)

14 November 2019 (X-Paper)
09:00–13:00

OPEN-BOOK EXAMINATION

Candidates may use textbooks and personal notes.

Calculators and mathematical tables may be used

No exchange of notes, books or communication between candidates will be allowed during the examination

This question paper consists of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
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NATIONAL CERTIFICATE
MECHANICAL DRAWING AND DESIGN N5
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Questions may be answered in any order, but subsections of questions must be kept together.
5. Show ALL intermediate the steps.
6. Sketches must be made FREEHAND and in good proportion in the ANSWER BOOK.
7. ALL the answers must be approximated to THREE decimal places.
8. Write neatly and legibly.

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QUESTION 1: WELDING

1.1 Obtain the throat area to be taken in the case of a 10 mm end fillet of 150 mm overall length if no return side fillets are used. (5)



1.2 If two 6 mm thick plates are to be welded with a butt joint, determine the required length of the weld if the joint is subjected to a tensile load of 44,5 kN. Take allowable tensile stress as 138 MPa (5)

[10]

QUESTION 2: LAP AND BUTT JOINTS

Two portions of a tie plate 50 mm wide and 12 mm thick are joined by a lap joint with one 20 mm diameter rivet. If the plates are subjected to a pulling force of 30 kN, find:

2.1 The tensile stress in the plates (3)

2.2 The shear stress in the rivet (3)



2.3 The crushing stress between the plate and the rivet (3)

2.4 Calculate the shear force on the rivet that will break the joint if the ultimate shear stress is 384 MPa. (3)

[12]

QUESTION 3: BELT DRIVES

A crossed flat belt drive is used to transmit power. The driving pulley is 320 mm in diameter and rotates at 680 revolutions per minute and the coefficient of friction is 0,4. The driven pulley is 600 mm in diameter and the centres of the pulleys are 900 mm apart. The drive belt can withstand a 7 kN tensile load and is 10 mm thick. The allowable tensile stress in the belt material is 4,5 MPa.

Calculate the following:

3.1 The contact angle of the driving pulley (3)

3.2 The width of the belt (2)



3.3 The total length of the belt (3)

3.4 The power transmitted by the belt (7)

[15]

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QUESTION 4: COTTER AND KNUCKLE JOINTS

Design a cotter joint to transmit 60 kN.
Use the following stresses:

Ultimate shear stress = 417 MN/m^2
Ultimate tensile stress = 460 MN/m^2
Ultimate crushing stress = 834 NM/m^2
Use a factor of safety of 5

Calculate the following:

4.1 Allowable stresses (3)

4.2 Diameter of pierced rod (2)

4.3 Breadth of cotter. (2)

4.4 Cotter thickness (5)

4.5 Diameter of rod (3)
[15]

QUESTION 5: SHAFTS

A solid shaft, 70 mm in diameter, transmits 125 kW at 440 r/min.

Calculate the following:

5.1 The torque transmitted (3)

5.2 The shear stress in the shaft (3)
[6]

QUESTION 6: COUPLINGS

A bushed-pin type flexible coupling for connecting an electric motor and a centrifugal pump shaft has to be designed.
The power to be transmitted is 18 kW at 1 000 r/min.
A pitch circle diameter of 152 mm for the pins will be used.
There are 6 pins of minimum diameter 12 mm, these values being chosen to keep the necessary dimensions of the rubber brushes reasonably low.

Calculate the following:

6.1 The torque transmitted (2)

6.2 The force acting on the pins (2)

6.3 The force on each pin (2)
[6]

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QUESTION 7: STEAM COVER

A rectangular steam chest opening, 350 mm by 310 mm, is to be closed by means of a flat cast-iron cover. The jointing material may be considered to extend to the bolts and to be subjected to the same pressure. The steam pressure is 1,5 MN/m^2 and the allowable stress for the bolts is 40 MPa. The steam chest walls are 23 mm thick. Use M22 bolts.

Calculate the following:

7.1	The effective area of the jointing material	(5)
7.2	The force exerted by the steam on the cover	(2)
7.3	The number of bolts if the core area = 0,8 nominal area	(3)
7.4	The pitch of the bolts 	(4)
7.5	Whether the cover will be steam tight	(2)
7.6	The total length and breadth of the cover	(4)

[20]

QUESTION 8: KEYS AND KEYWAYS

A feather key is 13 mm wide by 10 mm deep, and is to transmit 680 Nm of torque from a shaft 38 mm in diameter. The steel key has an allowable stress in tension and compression of 110 MPa and an allowable stress in shear of 55 MPa.

Determine the required length of the key.  **[6]**

QUESTION 9: BEARINGS

The two journals of a shaft carry a flywheel of mass 10 160 kg midway between them.
The coefficient of friction is 0,075 and the speed is 70 r/min.
The diameter of the journals is 300 mm.

Calculate the following:

9.1	Total load on bearings	(2)
9.2	Total friction	(2)
9.3	Distance moved in meters per minute	(2)
9.4	Work done per minute 	(2)
9.5	Power absorbed by friction	(2)

[10]

TOTAL: 100

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